higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N6

2 APRIL 2019

This marking guideline consists of 7 pages.

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QUESTION 1: COMPUTER SYSTEMS

1.1 1.1.1 B
1.1.2 B
1.1.3 D
1.1.4 C

(4 × 1) (4)

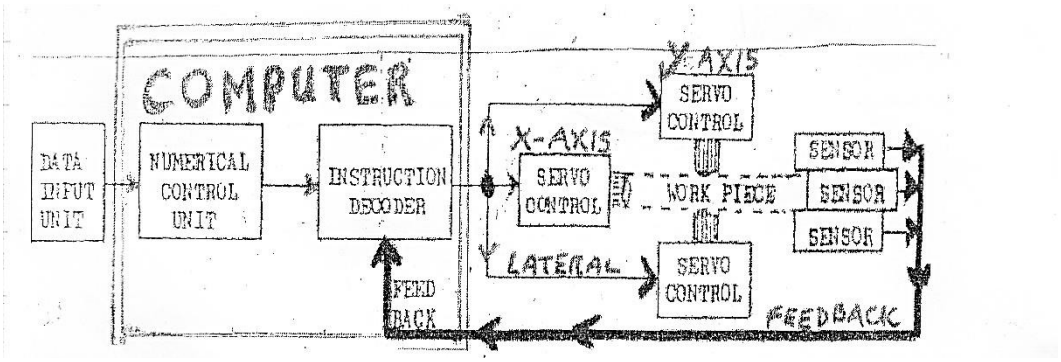
1.2 1.2.1 True
1.2.2 False
1.2.3 False
1.2.4 True

(4 × 1) (4)

1.3 1.3.1 C
1.3.2 F
1.3.3 A
1.3.4 D

(4 × 1) (4)

1.4



NOTE: Without the feedback loop the diagram is wrong. Including the feedback should have TWO marks awarded.

(a) Set points, data and modifications are entered at the data input unit.
(b) A cutting tool is moved sequentially and the feedback transmits a signal on completion after which the cutting tool is moved to the next position. (8)

[20]

QUESTION 2: TRANSMISSION, DATA ACQUISITION AND RELATED HARDWARE

2.1
2.1.1 A
2.1.2 C
2.1.3 D
2.1.4 C

(4 × 1) (4)

2.2
2.2.1 True
2.2.2 True
2.2.3 False
2.2.4 True

(4 × 1) (4)

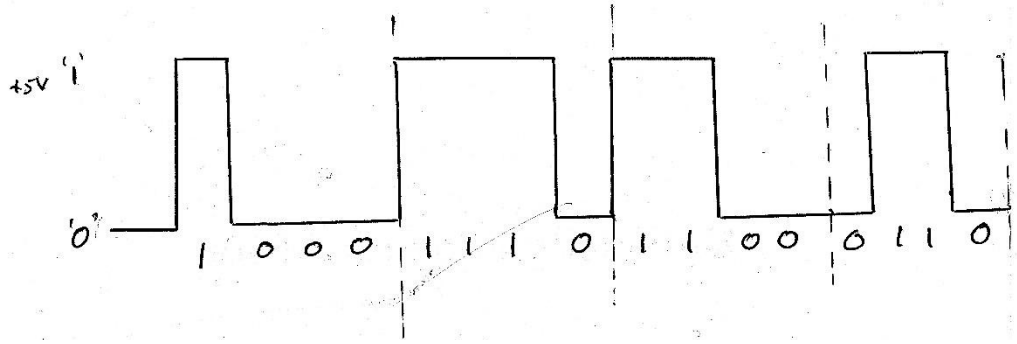
2.3
2.3.1 E
2.3.2 F
2.3.3 D
2.3.4 C

(4 × 1) (4)

2.4
- Data selection
- Data routing
- Operation sequencing
- Parallel-to-serial conversion
- Waveform generation
- Logic function generation

(Any 4 × 1) (4)

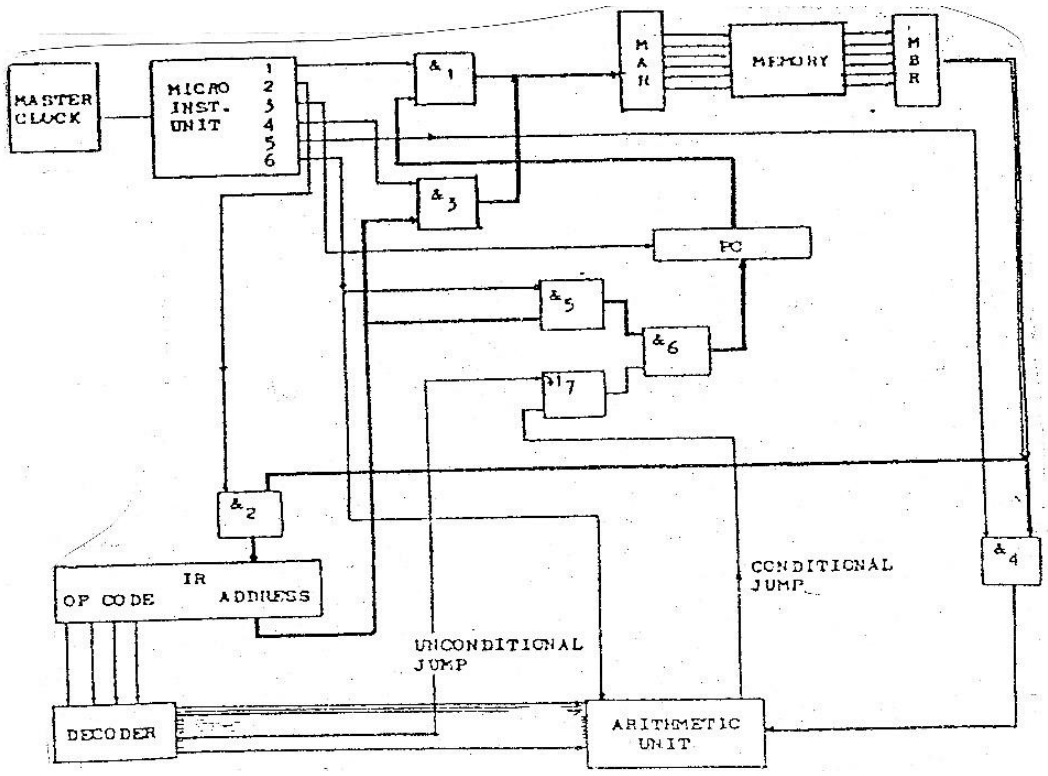
2.5



(4)
[20]

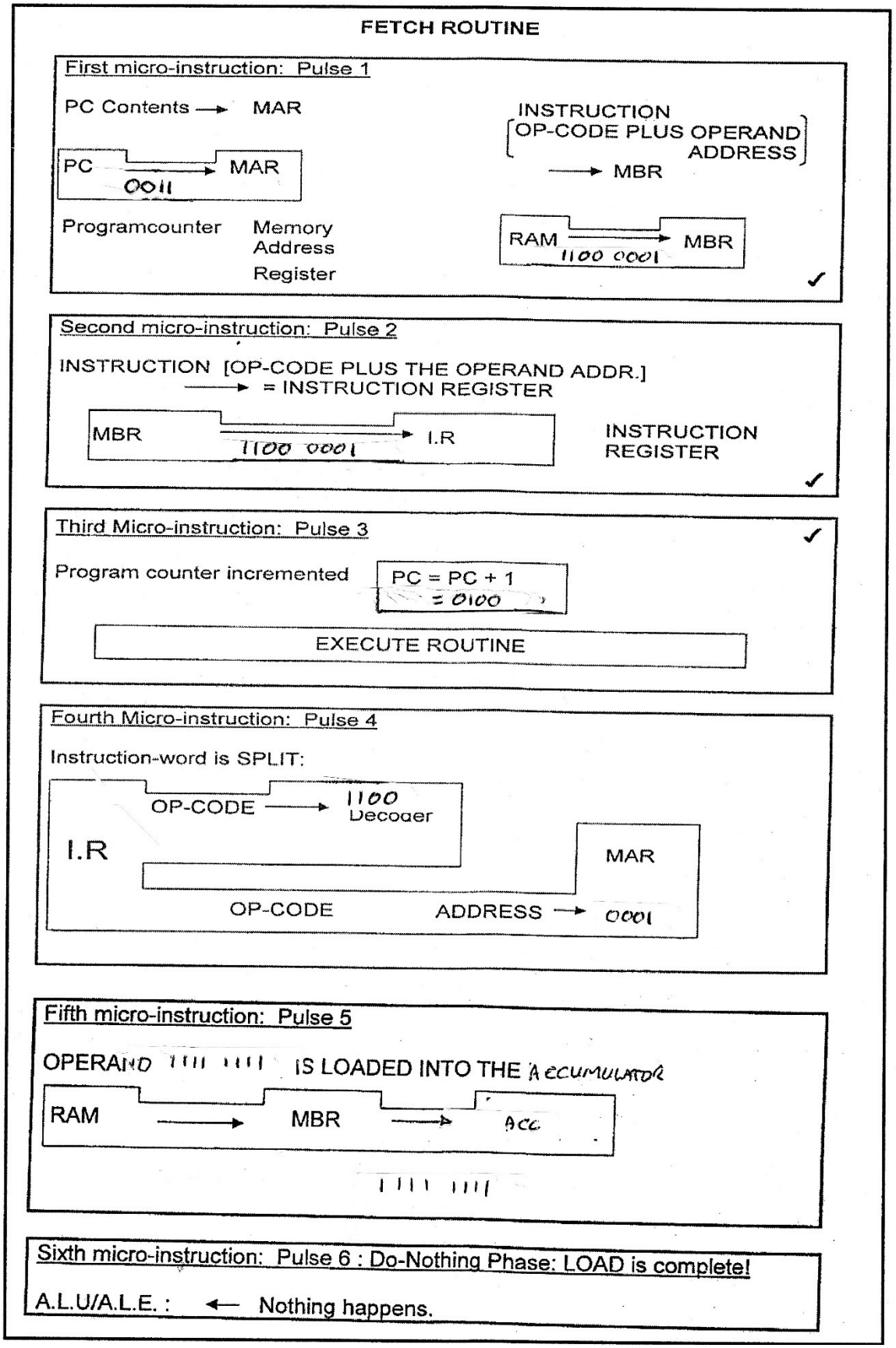
QUESTION 3: COMPUTER ARCHITECTURE

3.1



(10)

3.2


FETCH ROUTINE
First micro-instruction: Pulse 1
PC Contents → MAR
PC → MAR
0011
Programcounter
Memory Address Register
INSTRUCTION [OP-CODE PLUS OPERAND ADDRESS] → MBR
RAM → MBR
1100 0001
Second micro-instruction: Pulse 2
INSTRUCTION [OP-CODE PLUS THE OPERAND ADDR.] → = INSTRUCTION REGISTER
MBR → I.R
1100 0001
INSTRUCTION REGISTER
Third Micro-instruction: Pulse 3
Program counter incremented
PC = PC + 1
= 0100
EXECUTE ROUTINE
Fourth Micro-instruction: Pulse 4
Instruction-word is SPLIT:
OP-CODE → 1100 Decoder
I.R
MAR
OP-CODE
ADDRESS → 0001
Fifth micro-instruction: Pulse 5
OPERAND 1111 1111 IS LOADED INTO THE ACCUMULATOR
RAM → MBR → Acc
1111 1111
Sixth micro-instruction: Pulse 6 : Do-Nothing Phase: LOAD is complete!
A.L.U/A.L.E. : ← Nothing happens.

(10)
[20]

QUESTION 4: HIGH-LEVEL PROGRAMMING

4.1 THE WORDS OF WISDOM FOR TODAY ARE:

A SMILE CAN WORK WONDERS

NOTE: TWO marks if the top line is there and TWO spaces are left between it and the bottom line. EIGHT marks if the candidate could follow the program and not only get the printout correct but put it in capitals and all in one line (10)

4.2 Beginners' All-purpose Symbolic Instruction Code

NOTE: The candidate gets TWO marks or nothing. (2)

4.3
- Cost of a new system or expansion of an existing system
- Hiring of additional and specialised personnel
- Training of personnel
- Advantages and benefits derived from the proposed system
- Environmental considerations
- Problem areas and possible solutions
- Commissioning and installation
- Service and backup facilities
- Data files and format requirements at both input and output terminals
- Future expansion and estimated lifetime of the system
- Whether the cost of the end product surpasses the need (Any 6 × 1) (6)

4.4
- Real-time examples: airline, theatre ticket, bus or train reservations, ATM transactions
- Batch-processing examples: salaries and wages, billing for accounts (2)

[20]

QUESTION 5: NUMBER SYSTEMS

5.1 $1_1\ 0_2\ 0_3\ 0_4\ 0_5\ 1_6\ 1_7\ 1_8\ 1_9\ 0_{10}$

Pos. 1 checks 3; 5; 7; 9
0 0 1 1 – P1 should thus be 0: NOT thus: 1

Pos. 2 checks 3; 6; 7; 10
0 1 1 0 – P2 should thus be 0: IT IS thus: 0

Pos. 4 checks 5; 6; 7
0 1 1 – P4 should thus be 0: IT IS thus: 0

Pos. 8 checks 9; 10
1 0 – P8 should thus be 1: IT IS thus: 0

Thus the fault lies on bit $0001_2 – 1_{10}$

Thus pos. 1 which is a 1 should be a 0

i.e. the word should be: **$\underline{0}0000011110_{hamming}$** (10)

5.2 $+0{,}00011001 \times 10^{+110}$

$= 110{,}01_2$

$= 4 + 2 + 0{,}25$

$= 6{,}25_{10}$ (3)

5.3 $F = A.B.C + 0 + \bar{A}$
$= A.B.C + \bar{A}$
$= \bar{A} + A.B.C$
$= \bar{A} + B.C$

NOTE: Not ALL steps have to be shown. Some candidates who know Boolean algebra well should be able to get to the answer in one step. (3)

5.4 1110010_2

NOTE: ONE mark for the correct conversion, one for showing the subscript. (2)

5.5 The reflexive properties of the decimal equivalent (2)
[20]

TOTAL: 100

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